Database Schema (Banking System)

Create database Banking_System_054;

use Banking_System_054;

```
mysql> use Banking_System_054;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
```

show tables;

```
mysql> show tables;
+-------------------------------+
| Tables_in_Banking_System_054 |
+-------------------------------+
| Accounts_054                  |
| Branches                      |
| Branches_054                  |
| Customers_054                 |
| Loans_054                     |
| Transactions_054              |
+-------------------------------+
6 rows in set (0.00 sec)
```

<u>Customers</u>

customer_id (INT, PRIMARY KEY)

name (VARCHAR)

dob (DATE)

city (VARCHAR)

account_type (VARCHAR)

Describe Customers_054;

```
mysql> Describe Customers_054;
+--------------+--------------+------+-----+---------+-------+
| Field        | Type         | Null | Key | Default | Extra |
+--------------+--------------+------+-----+---------+-------+
| customer_id  | int          | NO   | PRI | NULL    |       |
| name         | varchar(100) | YES  |     | NULL    |       |
| DOB          | date         | YES  |     | NULL    |       |
| account_type | varchar(100) | YES  |     | NULL    |       |
| city         | varchar(100) | YES  |     | NULL    |       |
+--------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

select * from Customers_054;

```
mysql> select * from Customers_054;
+-------------+-------+------------+--------------+---------+
| customer_id | name  | DOB        | account_type | city    |
+-------------+-------+------------+--------------+---------+
|           1 | Rahul | 1998-05-10 | Pune         | Savings |
|           2 | Priya | 1997-08-21 | Mumbai       | Current |
| Google Chrome | Amit  | 1999-01-15 | Delhi        | Savings |
|           4 | Neha  | 1996-12-30 | Pune         | Savings |
|           5 | Arjun | 1995-07-18 | Mumbai       | Current |
+-------------+-------+------------+--------------+---------+
5 rows in set (0.00 sec)
```

<u>Accounts</u>

account_id (INT, PRIMARY KEY)

customer_id (INT, FOREIGN KEY REFERENCES Customers(customer_id))

balance (DECIMAL)

account_type (VARCHAR)

branch_id (INT, FOREIGN KEY REFERENCES Branches(branch_id))

Describe Accounts_054;

```
mysql> Describe Accounts_054;
+--------------+---------------+------+-----+---------+-------+
| Field        | Type          | Null | Key | Default | Extra |
+--------------+---------------+------+-----+---------+-------+
| account_id   | int           | NO   | PRI | NULL    |       |
| customer_id  | int           | YES  | MUL | NULL    |       |
| balance      | decimal(12,2) | YES  |     | NULL    |       |
| account_type | varchar(50)   | YES  |     | NULL    |       |
| branch_id    | int           | YES  | MUL | NULL    |       |
+--------------+---------------+------+-----+---------+-------+
5 rows in set (0.01 sec)
```

INSERT INTO Accounts_054 VALUES (1001, 1, 60000, 'Savings', 101), (1002, 2, 40000, 'Current', 102), (1003, 3, 20000, 'Savings', 103), (1004, 4, 15000, 'Savings', 101), (1005, 5, 80000, 'Current', 102), (1006, 1, 30000, 'Slavings', 102);

```
mysql> INSERT INTO Accounts_054 VALUES (1001, 1, 60000, 'Savings', 1
01), (1002, 2, 40000, 'Current', 102), (1003, 3, 20000, 'Savings', 1
03), (1004, 4, 15000, 'Savings', 101), (1005, 5, 80000, 'Current', 1
02), (1006, 1, 30000, 'Slavings', 102);
Query OK, 6 rows affected (0.02 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

select * from Accounts_054;

```
mysql> select * from Accounts_054;
+------------+-------------+----------+--------------+-----------+
| account_id | customer_id | balance  | account_type | branch_id |
+------------+-------------+----------+--------------+-----------+
|       1001 |           1 | 60000.00 | Savings      |       101 |
|       1002 |           2 | 40000.00 | Current      |       102 |
|       1003 |           3 | 20000.00 | Savings      |       103 |
|       1004 |           4 | 15000.00 | Savings      |       101 |
|       1005 |           5 | 80000.00 | Current      |       102 |
|       1006 |           1 | 30000.00 | Slavings     |       102 |
+------------+-------------+----------+--------------+-----------+
6 rows in set (0.00 sec)
```

<u>Transactions</u>

transaction_id (INT, PRIMARY KEY)

account_id (INT, FOREIGN KEY REFERENCES Accounts(account_id))

amount (DECIMAL)

transaction_type (VARCHAR) -- ('Deposit', 'Withdrawal',

'Transfer')

transaction_date (DATE)

Describe Transactions_054;

```
mysql> Describe Transactions_054;
+------------------+---------------+------+-----+---------+-------+
| Field            | Type          | Null | Key | Default | Extra |
+------------------+---------------+------+-----+---------+-------+
| transaction_id   | int           | NO   | PRI | NULL    |       |
| account_id       | int           | YES  | MUL | NULL    |       |
| amount           | decimal(12,2) | YES  |     | NULL    |       |
| transaction_type | varchar(20)   | YES  |     | NULL    |       |
| transaction_date | date          | YES  |     | NULL    |       |
+------------------+---------------+------+-----+---------+-------+
5 rows in set (0.01 sec)
```

INSERT INTO Transactions_054 VALUES (1, 1001, 10000, 'Deposit', '2025-12-01'), (2, 1001, 5000, 'Withdrawal', '2025-12-15'), (3, 1002, 20000, 'Deposit', '2025-11-10'), (4, 1003, 5000, 'Deposit', '2025-10-05'), (5, 1004, 2000, 'Withdrawal', '2025-09-01'), (6, 1005, 30000, 'Deposit', '2025-12-20'), (7, 1006, 10000, 'Transfer', '2025-12-28');

```
mysql> INSERT INTO Transactions_054 VALUES (1, 1001, 10000, 'Deposit', '2025-12-01'),
(2, 1001, 5000, 'Withdrawal', '2025-12-15'), (3, 1002, 20000, 'Deposit', '2025-11-10')
, (4, 1003, 5000, 'Deposit', '2025-10-05'), (5, 1004, 2000, 'Withdrawal', '2025-09-01'
), (6, 1005, 30000, 'Deposit', '2025-12-20'), (7, 1006, 10000, 'Transfer', '2025-12-28
');
Query OK, 7 rows affected (0.01 sec)
Records: 7  Duplicates: 0  Warnings: 0
```

select * from Transactions_054;

```
mysql> select * from Transactions_054;
+----------------+------------+----------+------------------+------------------+
| transaction_id | account_id | amount   | transaction_type | transaction_date |
+----------------+------------+----------+------------------+------------------+
|              1 |       1001 | 10000.00 | Deposit          | 2025-12-01       |
|              2 |       1001 |  5000.00 | Withdrawal       | 2025-12-15       |
|              3 |       1002 | 20000.00 | Deposit          | 2025-11-10       |
|              4 |       1003 |  5000.00 | Deposit          | 2025-10-05       |
|              5 |       1004 |  2000.00 | Withdrawal       | 2025-09-01       |
|              6 |       1005 | 30000.00 | Deposit          | 2025-12-20       |
|              7 |       1006 | 10000.00 | Transfer         | 2025-12-28       |
+----------------+------------+----------+------------------+------------------+
7 rows in set (0.00 sec)
```

<u>Branches</u>

branch_id (INT, PRIMARY KEY)

branch_name (VARCHAR)

city (VARCHAR)

Describe Branches_054;

```
mysql> Describe Branches_054;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| branch_id   | int          | NO   | PRI | NULL    |       |
| branch_name | varchar(100) | YES  |     | NULL    |       |
| city        | varchar(50)  | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

select * from Branches_054;

```
mysql> select * from Branches_054;
+-----------+-------------+--------+
| branch_id | branch_name | city   |
+-----------+-------------+--------+
|       101 | Pune_Main   | Pune   |
|       102 | Mumbai_Main | Mumbai |
|       103 | Delhi_Main  | Delhi  |
+-----------+-------------+--------+
3 rows in set (0.00 sec)
```

<u>Loans</u>

loan_id (INT, PRIMARY KEY)

customer_id (INT, FOREIGN KEY REFERENCES Customers(customer_id))

amount (DECIMAL)

loan_type (VARCHAR) -- ('Home', 'Car', 'Personal')

status (VARCHAR) -- ('Approved', 'Pending', 'Rejected')

Describe Loans_054;

```
mysql> Describe Loans_054;
+-------------+---------------+------+-----+---------+-------+
| Field       | Type          | Null | Key | Default | Extra |
+-------------+---------------+------+-----+---------+-------+
| loan_id     | int           | NO   | PRI | NULL    |       |
| customer_id | int           | YES  | MUL | NULL    |       |
| amount      | decimal(12,2) | YES  |     | NULL    |       |
| loan_type   | varchar(20)   | YES  |     | NULL    |       |
| status      | varchar(20)   | YES  |     | NULL    |       |
+-------------+---------------+------+-----+---------+-------+
5 rows in set (0.01 sec)
```

INSERT INTO Loans_054 VALUES (201, 1, 200000, 'Home', 'Approved'), (202, 2, 100000, 'Car', 'Pending'), (203, 3, 50000, 'Personal', 'Approved'), (204, 5, 150000, 'Home', 'Rejected');

```
mysql> INSERT INTO Loans_054 VALUES (201, 1, 200000, 'Home', 'Approved'), (202, 2, 100
000, 'Car', 'Pending'), (203, 3, 50000, 'Personal', 'Approved'), (204, 5, 150000, 'Hom
e', 'Rejected');
Query OK, 4 rows affected (0.02 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

select * from Loans_054;

```
mysql> select * from Loans_054;
+---------+-------------+-----------+-----------+----------+
| loan_id | customer_id | amount    | loan_type | status   |
+---------+-------------+-----------+-----------+----------+
|     201 |           1 | 200000.00 | Home      | Approved |
|     202 |           2 | 100000.00 | Car       | Pending  |
|     203 |           3 |  50000.00 | Personal  | Approved |
|     204 |           5 | 150000.00 | Home      | Rejected |
+---------+-------------+-----------+-----------+----------+
4 rows in set (0.00 sec)
```

<u>QUERIES :</u>

1. Retrieve the total balance for each account type.

SELECT account_type, SUM(balance) AS total_balance

-> FROM Accounts_054

-> GROUP BY account_type;

```
mysql> SELECT account_type, SUM(balance) AS total_balance
    -> FROM Accounts_054
    -> GROUP BY account_type;
+--------------+---------------+
| account_type | total_balance |
+--------------+---------------+
| Savings      |      95000.00 |
| Current      |     120000.00 |
| Slavings     |      30000.00 |
+--------------+---------------+
3 rows in set (0.01 sec)
```

2. Count the number of accounts in each branch.

SELECT branch_id, COUNT(*) AS total_accounts

-> FROM Accounts_054

-> GROUP BY branch_id;

```
mysql> SELECT branch_id, COUNT(*) AS total_accounts
    -> FROM Accounts_054
    -> GROUP BY branch_id;
+-----------+----------------+
| branch_id | total_accounts |
+-----------+----------------+
|       101 |              2 |
|       102 |              3 |
|       103 |              1 |
+-----------+----------------+
3 rows in set (0.00 sec)
```

3. Find the number of customers in each city.

SELECT city, COUNT(*) AS total_customers

-> FROM Customers_054

-> GROUP BY city;

```
mysql> SELECT city, COUNT(*) AS total_customers
    -> FROM Customers_054
    -> GROUP BY city;
+---------+-----------------+
| city    | total_customers |
+---------+-----------------+
| Savings |               3 |
| Current |               2 |
+---------+-----------------+
2 rows in set (0.00 sec)
```

4. Show the number of loans approved per loan type.

SELECT loan_type, COUNT(*) AS approved_loans

-> FROM Loans_054

-> WHERE status = 'Approved'

-> GROUP BY loan_type;

```
mysql> SELECT loan_type, COUNT(*) AS approved_loans
    -> FROM Loans_054
    -> WHERE status = 'Approved'
    -> GROUP BY loan_type;
+-----------+----------------+
| loan_type | approved_loans |
+-----------+----------------+
| Home      |              1 |
| Personal  |              1 |
+-----------+----------------+
2 rows in set (0.00 sec)
```

5. Find the total number of transactions for each transaction type.

SELECT transaction_type, COUNT(*) AS total_transactions FROM Transactions_054 GROUP BY transaction_type;

```
mysql> SELECT transaction_type, COUNT(*) AS total_transactions FROM Transactions_054 GROUP BY
transaction_type;
+------------------+--------------------+
| transaction_type | total_transactions |
+------------------+--------------------+
| Deposit          |                  4 |
| Withdrawal       |                  2 |
| Transfer         |                  1 |
+------------------+--------------------+
3 rows in set (0.00 sec)
```

6. Show the customers who do not have a loan.

SELECT *

-> FROM Customers_054

-> WHERE customer_id NOT IN (

-> SELECT customer_id FROM Loans_054

->);

```
mysql> SELECT *
    -> FROM Customers_054
    -> WHERE customer_id NOT IN (
    ->     SELECT customer_id FROM Loans_054
    -> );
+-------------+------+------------+--------------+---------+
| customer_id | name | DOB        | account_type | city    |
+-------------+------+------------+--------------+---------+
|           4 | Neha | 1996-12-30 | Pune         | Savings |
+-------------+------+------------+--------------+---------+
1 row in set (0.00 sec)
```

7. Retrieve customers who have accounts in more than one branch.

SELECT c.customer_id, c.name

-> FROM Customers_054 c

-> JOIN Accounts_054 a ON c.customer_id = a.customer_id

```
-> GROUP BY c.customer_id, c.name
-> HAVING COUNT(DISTINCT branch_id) > 1;
```

```
mysql> SELECT c.customer_id, c.name
    -> FROM Customers_054 c
    -> JOIN Accounts_054 a ON c.customer_id = a.customer_id
    -> GROUP BY c.customer_id, c.name
    -> HAVING COUNT(DISTINCT branch_id) > 1;
+-------------+-------+
| customer_id | name  |
+-------------+-------+
|           1 | Rahul |
+-------------+-------+
1 row in set (0.00 sec)
```

8. Find all accounts that have not had any deposits in the last 3 months.

```
SELECT account_id
    -> FROM Accounts_054
    -> WHERE account_id NOT IN (
    ->     SELECT account_id
    ->     FROM Transactions_054
    ->     WHERE transaction_type = 'Deposit'
    ->     AND transaction_date >= CURRENT_DATE - INTERVAL 3 MONTH
    -> );
```

```
mysql> SELECT account_id
    -> FROM Accounts_054
    -> WHERE account_id NOT IN (
    ->     SELECT account_id
    ->     FROM Transactions_054
    ->     WHERE transaction_type = 'Deposit'
    ->     AND transaction_date >= CURRENT_DATE - INTERVAL 3 MONTH
    -> );
+------------+
| account_id |
+------------+
|       1001 |
|       1006 |
|       1002 |
|       1003 |
|       1004 |
+------------+
5 rows in set (0.01 sec)
```

9. Display the account types where the total balance is below ₹25,000.

SELECT account_type, SUM(balance) AS total_balance

-> FROM Accounts_054

-> GROUP BY account_type

-> HAVING SUM(balance) < 25000;

```
mysql> SELECT account_type, SUM(balance) AS total_balance
    -> FROM Accounts_054
    -> GROUP BY account_type
    -> HAVING SUM(balance) < 25000;
Empty set (0.00 sec)
```

10. Find the total balance per account type where the balance is above ₹50,000.

SELECT account_type, SUM(balance) AS total_balance

-> FROM Accounts_054

-> WHERE balance > 50000

-> GROUP BY account_type;

```
mysql> SELECT account_type, SUM(balance) AS total_balance
    -> FROM Accounts_054
    -> WHERE balance > 50000
    -> GROUP BY account_type;
+--------------+---------------+
| account_type | total_balance |
+--------------+---------------+
| Savings      |      60000.00 |
| Current      |      80000.00 |
+--------------+---------------+
2 rows in set (0.00 sec)
```

11. Retrieve the count of transactions made on dates where more than 5 transactions Occurred.

SELECT transaction_date, COUNT(*) AS total_transactions

-> FROM Transactions_054

-> GROUP BY transaction_date

-> HAVING COUNT(*) > 5;

```
mysql> SELECT transaction_date, COUNT(*) AS total_transactions
    -> FROM Transactions_054
    -> GROUP BY transaction_date
    -> HAVING COUNT(*) > 5;
Empty set (0.00 sec)
```

12. Find the top 3 transaction days with the highest total transaction amount.

SELECT transaction_date, SUM(amount) AS total_amount

-> FROM Transactions_054

-> GROUP BY transaction_date

-> ORDER BY total_amount DESC

-> LIMIT 3;

```
mysql> SELECT transaction_date, SUM(amount) AS total_amount
    -> FROM Transactions_054
    -> GROUP BY transaction_date
    -> ORDER BY total_amount DESC
    -> LIMIT 3;
+------------------+--------------+
| transaction_date | total_amount |
+------------------+--------------+
| 2025-12-20       |     30000.00 |
| 2025-11-10       |     20000.00 |
| 2025-12-01       |     10000.00 |
+------------------+--------------+
3 rows in set (0.00 sec)
```

13. Find customers who have a loan but no account in the bank.

SELECT c.customer_id, c.name

-> FROM Customers_054 c

-> JOIN Loans_054 l ON c.customer_id = l.customer_id

-> LEFT JOIN Accounts_054 a ON c.customer_id = a.customer_id

-> WHERE a.account_id IS NULL;

```
mysql> SELECT c.customer_id, c.name
    -> FROM Customers_054 c
    -> JOIN Loans_054 l ON c.customer_id = l.customer_id
    -> LEFT JOIN Accounts_054 a ON c.customer_id = a.customer_id
    -> WHERE a.account_id IS NULL;
Empty set (0.00 sec)
```